

Electricity foundations — Class 10

Original Class 10 notes — current, circuits, graphs and resistance

Detailed foundations with nine original checked worked examples and source-verified Activities 11.1–11.6 from NCERT Chapter 11, Reprint 2026–27.

Scope and source

These original notes explain current, potential difference, resistance, measurements, graphs, wire geometry and resistor networks. All worked examples are newly written teaching examples, not official NCERT or CBSE questions. They make no predictions or guarantees about examinations. Heating, power and examination guidance belong to the companion document.

The reference is NCERT Science, Chapter 11, Electricity, Reprint 2026–27. Citations below use printed textbook pages, not PDF page numbers. The official source throughout is <https://ncert.nic.in/textbook/pdf/jesc111.pdf>. Explanations and examples are original; activity descriptions are paraphrased. Numerical circuits assume ideal connecting wires and meters, steady current and constant resistor temperature unless stated otherwise.

1. Current: charge crossing a section

Electric current measures how quickly charge passes through a cross-section of a conductor: $I = Q/t$. Thus $Q = It$ and $t = Q/I$. For steady current this ratio applies throughout the interval; over an arbitrary interval it gives the average current. Charge is measured in coulombs, C, and time in seconds, s. One ampere means one coulomb passing per second: $1 \text{ A} = 1 \text{ C/s}$.

A closed conducting path is needed for sustained current in a simple cell-and-resistor circuit. Opening its only path stops current. In metallic wires electrons move opposite to conventional current. Through the external circuit, conventional current runs from the cell's positive terminal towards its negative terminal. Current is not the number of electrons sitting inside a wire.

Source: NCERT section 11.1, printed pp. 171–172; <https://ncert.nic.in/textbook/pdf/jesc111.pdf>.

2. Potential difference: work per charge

Potential difference describes work done per unit charge between two points: $V = W/Q$, so $W = VQ$. One volt means one joule per coulomb: $1 \text{ V} = 1 \text{ J/C}$. A 3 V potential difference corresponds to 3 J for every coulomb transferred between the specified points, not 3 C every second.

A cell's chemical action maintains a potential difference. Its terminals can have a potential difference even when the external circuit is open and current is zero. Therefore voltage and current are different quantities. Always identify the two points meant by a voltage; saying “voltage through the resistor” confuses an across-two-points measurement with current through a component.

Source: NCERT section 11.2, printed pp. 173–174; <https://ncert.nic.in/textbook/pdf/jesc111.pdf>.

3. Units and conversions before substitution

Convert data before combining formulas. Keep the symbol A for ampere distinct from A used for cross-sectional area; context and units identify the meaning.

Source: NCERT sections 11.1, 11.4 and 11.5, printed pp. 172, 176, 178–180; conversions apply standard SI prefixes. <https://ncert.nic.in/textbook/pdf/jesc111.pdf>.

- $1 \text{ mA} = 10^{-3} \text{ A}$; $1 \text{ }\mu\text{A} = 10^{-6} \text{ A}$; $750 \text{ mA} = 0.750 \text{ A}$.
- $1 \text{ k}\Omega = 10^3 \Omega$; $1 \text{ M}\Omega = 10^6 \Omega$; $2.4 \text{ k}\Omega = 2400 \Omega$.

- 1 minute = 60 s; 2.5 minutes = 150 s.
- 1 cm = 10^{-2} m and 1 mm = 10^{-3} m.
- 1 mm² = 10^{-6} m², not 10^{-3} m²: an area conversion squares the length factor.
- Resistance uses Ω ; resistivity uses Ω m. Neither unit can replace the other.

4. Meters and circuit interpretation

Insert an ammeter in series with the path whose current is required. Connect a voltmeter in parallel across the two endpoints being compared. An ideal ammeter adds no resistance; an ideal voltmeter draws no current. These are calculation assumptions, not literal descriptions of real instruments. Never connect an ammeter directly across a cell.

Follow connections rather than the shape of the drawing. A junction divides a path; wires crossing without joining do not form a junction. Two resistors are in series only when the same current must pass through both without branching between them. Parallel resistors share both endpoints, even if they look far apart on paper. A bend in an ideal wire does not create a voltage drop.

For supervised low-voltage measurements, open the key before changing connections, check meter polarity and range, and read scales without parallax. Determine the value per division from the marked scale. Use appropriately rated apparatus; these notes do not instruct experiments with mains electricity.

Source: NCERT sections 11.1–11.3, printed pp. 172–175; Activities 11.2 and 11.6, pp. 176–177 and 185–186. Meter idealisations and handling reminders are explanatory additions.
<https://ncert.nic.in/textbook/pdf/jesc111.pdf>.

5. Resistance and the condition in Ohm's law

Resistance describes opposition to charge flow. $R = V/I$, measured in ohms: a conductor has resistance 1 Ω when 1 V across it produces 1 A through it. For a given metallic conductor at constant temperature, Ohm's law states $V \propto I$, giving $V = IR$ with constant R .

Writing V/I for one observation does not prove that resistance stays constant over other observations. Test several pairs while keeping temperature unchanged. A wire that warms appreciably may give changing ratios. At fixed V , doubling R halves I ; at fixed R , doubling V doubles I . State what remains fixed before drawing either conclusion. A rheostat changes circuit resistance to regulate current without changing the voltage source.

Source: NCERT section 11.4, printed pp. 175–177; temperature dependence also section 11.5, p. 179. <https://ncert.nic.in/textbook/pdf/jesc111.pdf>.

6. Both graph orientations: read the axes first

An ohmic conductor at constant temperature gives a straight line through the origin. "Steeper means greater resistance" is incomplete unless the axes and their units are known. Compare slopes on identical numerical scales, not merely the apparent angle of separately drawn graphs.

Source: NCERT section 11.4, Activity 11.1 and Fig. 11.3, printed pp. 175–176. Reciprocal-axis slopes follow algebraically from $V = IR$. <https://ncert.nic.in/textbook/pdf/jesc111.pdf>.

- V–I graph: horizontal axis I in A; vertical axis V in V. Slope = $\Delta V/\Delta I = R$, in Ω . A steeper line means larger resistance.
- I–V graph: horizontal axis V in V; vertical axis I in A. Slope = $\Delta I/\Delta V = 1/R$, in A/V. A steeper line means smaller resistance.
- Use two widely separated points on a best-fit straight line, subtract coordinates in matching order, and convert mA to A before reporting resistance in Ω .

- Small experimental scatter is not automatically failure of Ohm's law. Check readings and temperature; do not force a clearly curved relationship into a constant-resistance interpretation.

7. Resistivity, area and diameter

For a uniform wire, $R = \rho l/A$. Resistance depends on length l , cross-sectional area A , material and temperature. Resistivity ρ describes the material at a specified temperature; changing length alone does not change ρ . Rearranging gives $\rho = RA/l$, whose unit is $\Omega \text{ m}$.

For the same material at the same temperature, $R_2/R_1 = (l_2/l_1)(A_1/A_2)$. Longer wires resist more when area is fixed; thicker wires resist less when length is fixed. For a circular section, $A = \pi d^2/4$. Therefore doubling diameter quadruples area and quarters resistance at unchanged length. Doubling area only halves resistance. "Twice as thick" is ambiguous: specify diameter or area.

Source: NCERT section 11.5, printed pp. 177–180, including Eq. 11.10 and the diameter calculation in Example 11.5. <https://ncert.nic.in/textbook/pdf/jesc111.pdf>.

8. Stretching a wire: two changes together

For uniform stretching, explicitly assume constant volume, unchanged material resistivity and constant temperature. If length becomes n times its original value, $lA = l'A'$ gives $A' = A/n$. Substitution then gives $R' = \rho(nl)/(A/n) = n^2R$. For a circular wire, $d' = d/\sqrt{n}$.

Do not use $R' = nR$: that result holds when area stays fixed, whereas stretching makes the wire thinner. This is a derived application of the NCERT geometry relation, not a claim that every real stretching process preserves resistivity. Cutting a wire into equal pieces is different: each piece keeps the original area but has a shorter length.

Source basis: NCERT section 11.5, Eq. 11.10, printed p. 178; ratio reasoning on p. 180. Constant-volume stretching is an original extension. <https://ncert.nic.in/textbook/pdf/jesc111.pdf>.

9. Series resistance: derive before using

Series resistors carry the same current I . The total potential difference equals the sum of individual drops: $V = V_1 + V_2 + V_3$. Applying Ohm's law gives $IR_s = IR_1 + IR_2 + IR_3$; cancelling nonzero I gives $R_s = R_1 + R_2 + R_3$.

For positive resistances, the result exceeds every individual resistance. Since $V_1 = IR_1$ and $V_2 = IR_2$, $V_1/V_2 = R_1/R_2$: the larger resistor receives the larger voltage drop. Equal current does not mean equal voltage. Breaking the only series path stops current throughout it.

Source: NCERT section 11.6.1, Activities 11.4–11.5 and Eqs. 11.11–11.14, printed pp. 182–184; broken-path discussion p. 187. <https://ncert.nic.in/textbook/pdf/jesc111.pdf>.

10. Parallel resistance and mixed networks

Parallel branches share the same potential difference V . The total current is $I = I_1 + I_2 + I_3$. Substituting $I = V/R$ gives $V/R_p = V/R_1 + V/R_2 + V/R_3$; cancelling nonzero V gives $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$.

For two resistors only, $R_p = R_1R_2/(R_1 + R_2)$. For n identical resistors, $R_p = R/n$. Equivalent resistance is smaller than the smallest positive branch resistance. Since $I_1/I_2 = R_2/R_1$, the lower-resistance branch carries more current; current does not divide equally unless branch resistances are equal.

In mixed networks, label junctions and reduce one genuine series or parallel group at a time. Find total current, then work backwards to group voltages and branch currents. Do not assign the whole battery voltage to a parallel group that has another resistor in series with it.

Source: NCERT section 11.6.2, printed pp. 185–187, Eqs. 11.15–11.18 and mixed-network Example 11.9. <https://ncert.nic.in/textbook/pdf/jesc111.pdf>.

11. Original worked examples: charge and voltage

Example 1 — Charge transfer. A steady current of 240 mA flows for 2.5 minutes. Convert $I = 0.240$ A and $t = 150$ s. Then $Q = It = 0.240 \times 150 = 36$ C. Check: $36/150 = 0.240$ C/s, recovering the current. Using 2.5 without converting minutes would give the wrong charge.

Example 2 — Work per coulomb. Transferring 0.75 C between two points requires 4.5 J. $V = W/Q = 4.5/0.75 = 6$ V. For 2 C between the same points, $W = VQ = 6 \times 2 = 12$ J. Check: $12/2 = 6$ J/C. More transferred charge increases total work, not the fixed potential difference.

Formula sources: NCERT sections 11.1–11.2, printed pp. 172–174; <https://ncert.nic.in/textbook/pdf/jesc111.pdf>.

12. Original worked examples: resistance and graph slopes

Example 3 — Milliampere conversion. A resistor carries 4 mA at 6 V. $I = 0.004$ A, so $R = 6/0.004 = 1500 \Omega = 1.5$ k Ω . At 3 V and unchanged temperature, $I = 3/1500 = 0.002$ A = 2 mA. Check: $IR = 0.002 \times 1500 = 3$ V.

Example 4 — One resistor, two plots. Points on a straight line are $(I, V) = (0.10$ A, 1.5 V) and $(0.30$ A, 4.5 V). With V vertical, slope = $(4.5 - 1.5)/(0.30 - 0.10) = 15 \Omega$. With I vertical, slope = $0.20/3.0 = 1/15$ A/V; its reciprocal is 15Ω . Check: both pointwise V/I ratios equal 15Ω .

Formula source: NCERT section 11.4, printed pp. 175–176; <https://ncert.nic.in/textbook/pdf/jesc111.pdf>.

13. Original worked examples: wire geometry

Example 5 — Find resistivity. A uniform wire has $R = 3 \Omega$, $l = 1.5$ m and $A = 0.20$ mm² = 2.0×10^{-7} m². Thus $\rho = RA/l = (3 \times 2.0 \times 10^{-7})/1.5 = 4.0 \times 10^{-7} \Omega$ m. Check: $\rho l/A$ returns 3Ω . This hypothetical value is not assigned to a named material.

Example 6 — Diameter versus stretching. A wire initially has resistance 12Ω . A separate wire of the same material, temperature and length but twice its diameter has resistance $12/4 = 3 \Omega$. Instead, stretching the original uniformly to 1.5 times its length at constant volume, temperature and resistivity gives $R' = 1.5^2 \times 12 = 27 \Omega$. Check: $A' = A/1.5$, so the geometry multiplier is $1.5/(1/1.5) = 2.25$.

Formula source: NCERT section 11.5, printed pp. 178–180; <https://ncert.nic.in/textbook/pdf/jesc111.pdf>.

14. Original worked examples: complete networks

Example 7 — Series. Connect 4Ω and 8Ω in series across 6 V. $R_s = 12 \Omega$ and $I = 6/12 = 0.5$ A through each resistor. Voltage drops are $0.5 \times 4 = 2$ V and $0.5 \times 8 = 4$ V. Check: $2 + 4 = 6$ V and $2:4 = 4:8$.

Example 8 — Parallel. Connect 8Ω and 24Ω across 6 V. $R_p = (8 \times 24)/(8 + 24) = 6 \Omega$. Branch currents are $6/8 = 0.75$ A and $6/24 = 0.25$ A. Total current = 1 A. Check: $V/I = 6 \Omega$, below 8Ω , and branch currents sum correctly.

Example 9 — Mixed. A 3Ω resistor is in series with a parallel pair of 6Ω and 12Ω across 7 V. The pair equals $72/18 = 4 \Omega$; total resistance is 7Ω and total current is 1 A. The series resistor drops 3 V, leaving 4 V across both branches. Branch currents are $4/6 = 2/3$ A and $4/12 = 1/3$ A. Check: branch currents sum to 1 A and voltage drops sum to 7 V.

Formula sources: NCERT sections 11.6.1–11.6.2, printed pp. 182–187;
<https://ncert.nic.in/textbook/pdf/jesc111.pdf>.

15. Activities 11.1–11.6: procedure, observation, meaning

These are paraphrased source-based expectations, not measurements performed for these notes. Actual readings depend on apparatus and uncertainty. Change connections with the key open and use teacher-supervised low-voltage apparatus.

Official reference for every activity: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>.

- Activity 11.1, section 11.4, p. 175; interpretation p. 176: measure I through and V across a nichrome wire using one, two, three and four 1.5 V cells successively. V/I should be approximately constant and the graph straight through the origin when temperature stays constant. Record measured voltage, not just nominal cell totals.
- Activity 11.2, section 11.4, pp. 176–177: replace the component in gap XY successively with nichrome wire, a torch bulb and the listed 10 W bulb, using four dry cells. Different current readings indicate differing opposition to current. Do not invent a universal ranking without component data. Remove the key after each reading.
- Activity 11.3, section 11.5, pp. 177–178: compare the original nichrome wire with twice its length, a thicker wire of the original length, and copper of matching length and area. NCERT reports half current on doubling length, increased current with thickness, and a change with material. The half-current comparison assumes essentially unchanged voltage across the test wire.
- Activity 11.4, section 11.6.1, p. 182: move the ammeter between positions in a three-resistor series path. The reading remains the same: current is common throughout the series combination, not consumed by successive resistors.
- Activity 11.5, section 11.6.1, p. 183: measure voltage across the whole series combination, compare it with battery terminal voltage, then measure each resistor separately. Observe $V = V_1 + V_2 + V_3$; unequal resistors need not have equal drops.
- Activity 11.6, section 11.6.2, pp. 185–186: measure total current and voltage for three parallel resistors; then measure each branch current with the ammeter in that branch. Voltage is common and $I = I_1 + I_2 + I_3$, not necessarily three equal currents.

16. Misconceptions to replace

Source basis: NCERT sections 11.1–11.6, printed pp. 171–187, especially pp. 173, 176, 178–179, 182–186; <https://ncert.nic.in/textbook/pdf/jesc111.pdf>.

- “Current gets used up.” Charge continues through a steady series path; current is unchanged between resistors. Energy transfer is a different idea.
- “A battery always supplies the same current.” Current depends on the connected resistance as well as terminal voltage.
- “Zero current means zero voltage everywhere.” An open cell-powered circuit can retain a terminal potential difference.
- “A thick wire has lower resistivity.” Geometry changes resistance; resistivity belongs to the material at its temperature.
- “Resistance always equals graph slope.” Only a graph with V vertical and I horizontal has slope R .
- “Adding resistors always increases resistance.” Adding a series resistor increases it; adding a positive parallel branch decreases it.

Official curriculum scope and assessment limitation

CBSE Science (086), Class X, 2026-27, PDF p.6 Unit IV lists electric current, potential difference, resistance, resistivity, Ohm's law, resistor combinations, heating and power. Practical 4 on PDF p.7 covers V-I measurements; practical 5 on PDF p.8 covers series and parallel equivalent resistance. Source: https://cbseacademic.nic.in/web_material/CurriculumMain27/SecPart1/Science_SecP1_2026-27.pdf

The same PDF p.6 contains an unresolved teacher-note wording inconsistency naming "Electric Effects of Electric Current" among unassessed topics, despite the explicit Unit IV list. These are chapter study materials, not a guarantee of summative assessment coverage; consult school/CBSE clarification. Unit IV's 13 marks covers a combined unit, not an Electricity-only allotment.